Topic : Pre-emptive scheduling algorithm and their types (SRTF)

Objective : The main purposes of scheduling algorithms are to minimize resource starvation and to ensure fairness amongst the parties utilizing the resources.

Outcome : Students will learn how process switches from running state to ready state or from the waiting state to ready state.

PRE-EMPTIVE SCHEDULING ALGORITHM

Preemptive scheduling is used when a process switches from running state to ready state or from the waiting state to ready state. The resources (mainly CPU cycles) are allocated to the process for a limited amount of time and then taken away, and the process is again placed back in the ready queue if that process still has CPU burst time remaining. That process stays in the ready queue till it gets its next chance to execute.

the preemptive scheduling is based on priority where a scheduler may preempt a low priority running process anytime when a high priority process enters into a ready state.

Preemptive Scheduling is a CPU scheduling technique that works by dividing time slots of CPU to a given process. The time slot given might be able to complete the whole process or might not be able to it. When the burst time of the process is greater than CPU cycle, it is placed back into the ready queue and will execute in the next chance. This scheduling is used when the process switch to ready state.

SRTF (Shortest Remaining Time First)

- Preemptive mode of Shortest Job First is called as **Shortest Remaining Time First (SRTF)**.

The SRTF algorithm involves more overheads than the Shortest job first (SJF)scheduling, because in SRTF OS is required frequently in order to monitor the CPU time of the jobs in the **READY** queue and to perform context switching.

In the **SRTF scheduling algorithm**, the execution of any process can be stopped after a certain amount of time. On arrival of every process, the short-term scheduler schedules those processes from the list of available processes & running processes that have the least remaining burst time.

Advantages of SRTF

- The main advantage of the SRTF algorithm is that it makes the processing of the jobs faster than the SJF algorithm, mentioned it's overhead charges are not counted.
- SRTF is optimal and guarantees the minimum average waiting time.
- It provides a standard for other algorithms since no other algorithm performs better than it.

Disadvantages of SRTF

- In SRTF, the context switching is done a lot more times than in SJN due to more consumption of the CPU's valuable time for processing. The consumed time of CPU then adds up to its processing time and which then diminishes the advantage of fast processing of this algorithm.

Example

Summary : Shortest remaining time, also known as shortest remaining time first (SRTF), is a scheduling method that is a preemptive version of shortest job next scheduling. In this scheduling algorithm, the process with the smallest amount of time remaining until completion is selected to execute.

Topic : LRTF, Round Robin and Priority based scheduling algorithm

Objective : The main purposes of scheduling algorithms are to minimize resource starvation and to ensure fairness amongst the parties utilizing the resources.

Outcome : Students will learn how process switches from running state to ready state or from the waiting state to ready state.

LONGEST REMAINING TIME FIRST (LRTF)

LRTF stands for **Longest Remaining Time First**. In the LRTF scheduling algorithm, the job that has the highest burst time is allocated CPU first. We keep monitoring the burst times periodically and assign the then highest burst time process to the CPU. It is a preemptive mode of the Longest Job First algorithm.

Characteristics

- Longest Remaining Time First is a CPU scheduling algorithm used to determine the process to be executed first among all the incoming processes in a systematic way.
- It follows the preemptive approach i.e. CPU is allocated only for a fixed slice of time.
- The criteria for process selection is highest burst size and this process runs till the fixed slice. The selection process takes place again.
- This is not an optimal scheduling algorithm as the average waiting time is high.

Advantages

- LRTF algorithm is simple and easy to implement.
- Almost all the processes complete by the time the longest job reaches its completion.
- Starvation-free as all processes get fair share of CPU.

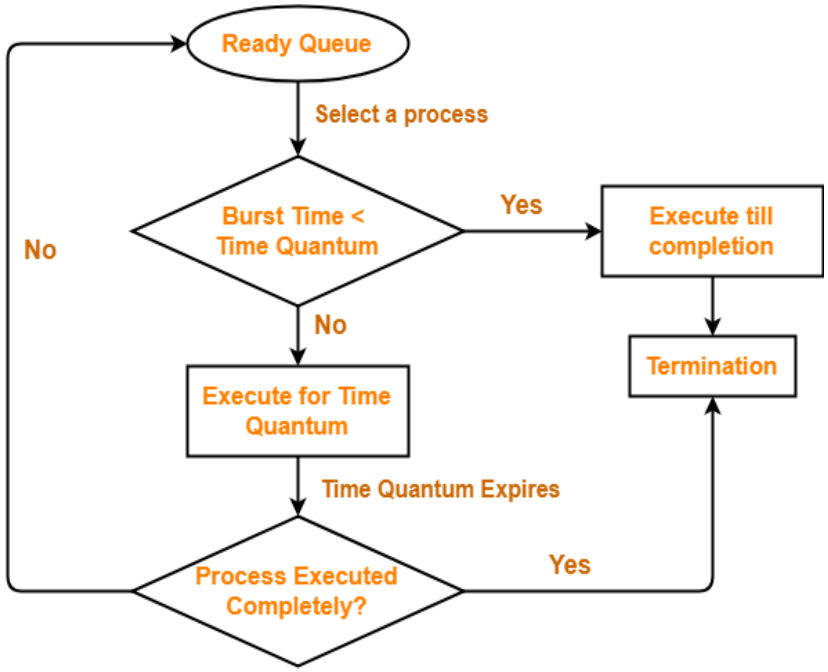
Disadvantages

- The context switch consumes CPU's valuable time which can be utilized for executing processes.
- Smaller processes need to wait for CPU to finish larger burst size processes.
- The average waiting time and average turn-around time are much high even with much less burst time of each process.

ROUND ROBIN SCHEDULING ALGORITHM

In Round Robin Scheduling,

- CPU is assigned to the process on the basis of FCFS for a fixed amount of time.
- This fixed amount of time is called as **time quantum** or **time slice**.
- After the time quantum expires, the running process is preempted and sent to the ready queue.
- Then, the processor is assigned to the next arrived process.
- It is always preemptive in nature.



Round Robin Scheduling

Advantages-

- It gives the best performance in terms of average response time.
- It is best suited for time sharing system, client server architecture and interactive system.

Disadvantages-

- It leads to starvation for processes with larger burst time as they have to repeat the cycle many times.
- Its performance heavily depends on time quantum.
- Priorities can not be set for the processes.

Important Notes-

Note-01:

With decreasing value of time quantum,

- Number of context switch increases
- Response time decreases
- Chances of starvation decreases

Note-02:

With increasing value of time quantum,

- Number of context switch decreases
- Response time increases
- Chances of starvation increases

Note-03:

- With increasing value of time quantum, Round Robin Scheduling tends to become FCFS Scheduling.
- When time quantum tends to infinity, Round Robin Scheduling becomes FCFS Scheduling.

Note-04:

- The performance of Round Robin scheduling heavily depends on the value of time quantum.
- The value of time quantum should be such that it is neither too big nor too small.

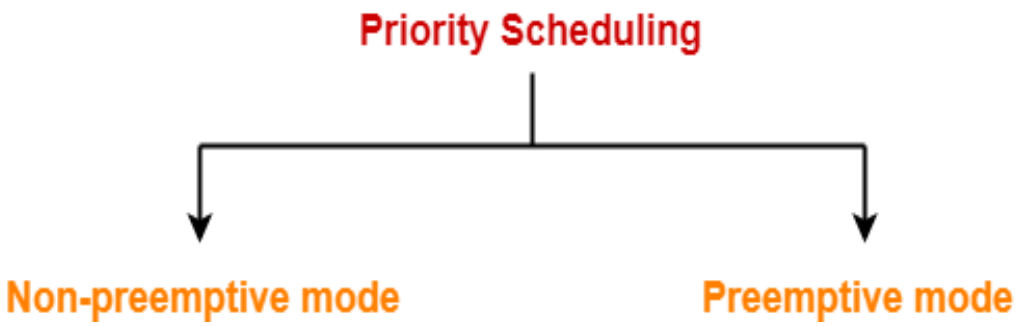
PRIORITY BASED SCHEDULING ALGORITH

Priority Scheduling is a method of scheduling processes that is based on priority. In this algorithm, the scheduler selects the tasks to work as per the priority.

The processes with higher priority should be carried out first, whereas jobs with equal priorities are carried out on a round-robin or FCFS basis. Priority depends upon memory requirements, time requirements, etc.

In Priority Scheduling,

- Out of all the available processes, CPU is assigned to the process having the highest priority.
- In case of a tie, it is broken by FCFS Scheduling.
- Priority Scheduling can be used in both preemptive and non-preemptive mode.



Characteristics of Priority Scheduling

- A CPU algorithm that schedules processes based on priority.
- It used in Operating systems for performing batch processes.
- If two jobs having the same priority are READY, it works on a FIRST COME, FIRST SERVED basis.
- In priority scheduling, a number is assigned to each process that indicates its priority level.
- Lower the number, higher is the priority.
- In this type of scheduling algorithm, if a newer process arrives, that is having a higher priority than the currently running process, then the currently running process is preempted.

Advantages-

- It considers the priority of the processes and allows the important processes to run first.
- Priority scheduling in preemptive mode is best suited for real time operating system.

Disadvantages-

- Processes with lesser priority may starve for CPU.
- There is no idea of response time and waiting time.

Important Notes-

Note-01:

- The waiting time for the process having the highest priority will always be zero in preemptive mode.
- The waiting time for the process having the highest priority may not be zero in non-preemptive mode.

Note-02:

Priority scheduling in preemptive and non-preemptive mode behaves exactly same under following conditions-

- The arrival time of all the processes is same
- All the processes become available

Summary - Its main objective is **to increase system performance in accordance with the chosen set of criteria**. It is the change of ready state to running state of the process. CPU scheduler selects a process among the processes that are ready to execute and allocates CPU to one of them.